

Lecture	T/Th 5:45-7:45, UW1-220
Instructor	Prof Pisan Email: pisan@uw.edu
Office	UW1-260Q Phone: 425.352.3741
Office Hours	W 2-3pm, Th 2-3pm or by appointment

Course Description

Introduces programming concepts within social, mathematical, and technological context. Topics include programming fundamentals (control structures, data types, functions, etc.), computer organization, algorithmic thinking, introductory software engineering concepts, and social and professional issues. Engineering applications are emphasized. A computer language used in engineering practice is used for instruction. Co-requisite: CSSSKL 132.

Learning Objectives

- Develop competencies associated with problem-solving, design, programming, and testing techniques;
- See common applications and consider these applications in society;
- Learn and use good software engineering and algorithm analysis techniques; and
- Become proficient in using C/C++ for programming.

Textbooks

zyBooks: *Programming in C++*, F Vahid and R Lysecky,
<https://www.zybooks.com/catalog/programming-c-plus-plus/>

This is an e-book with integrated exercises. To get the e-book:

1. Sign in or create an account at <https://learn.zybooks.com/> with your @uw.edu email
2. Enter zyBook code: UWBCSS132PisanFall2019
3. Subscribe

For additional books and references see <https://github.com/pisanorg/w/wiki>

Grading

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide.
2.0 roughly corresponds to 75%

Participation: 10% -- See P01-P10 in zybooks

Homework: 25% -- See H01-H10 in zybook

3 Semester Exams: 40% -- (10+15+15) see dates below

Final Exam: 25%

For Participation and Homework, points are converted as follows:

90-100% of the points - 100

80-89% - 90

70-79% - 80

0-69% - 0

You are required to enroll and attend CSSSKL 132 which is separately graded as Credit/No Credit.

Policies

Attendance: Attend all classes. If you are going to miss a class, let me know via email. If you miss a class, contact your classmates to find out what you missed. If you miss more than one class, come to my office hours to explain your circumstances.

Electronics: Minimal electronics in class unless we have a specific class exercise that requires a computer. Research shows that using pen and paper to take notes is much more effective than using a computer. Computers are often distracting to you and also to others around you. If you need accommodations for taking notes, let me know. If you need to send a text message or take a phone call, step outside to do it.

Assignment Submission: Strict deadlines, but everybody occasionally slips, so you can submit 1 assignment up to 24 hours late for any reason (email me to let me know). All other assignments have to be submitted on time. If you need an extension due to exceptional circumstances, email or talk to me before the due date. Exceptional circumstances only; work, other classes, sleep, traffic, etc are not exceptional.

Effort: Expect to spend 10-15 hours per week outside class. If an assignment is worth 10%, expect to spend 10-20 hours on the assignment. If you are spending too much time or too little time, let me know we'll adjust the course content. Learning happens best when you are challenged and get to stretch your limits.

Academic Integrity: Do the right thing, see <http://guides.lib.uw.edu/bothell/ai> for details on what is "right" if in doubt. Talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK.

Communication: Join the course channel on discord <https://discord.gg/5mEm92e> . Discord is an extension of the classroom, so use a meaningful nickname and act professionally. If your question can be answered by a classmate, post it to discord rather than using email or waiting for office hours. Office hours are for complex issues or topics you are struggling with.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Access and Accommodations: Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course. See <http://www.uwb.edu/studentaffairs/drs> if you need to establish accommodations.

Religious Accommodations: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Faculty Syllabus Guidelines and Resources. Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form available at <https://registrar.washington.edu/students/religious-accommodations-request/>

Common Course Policies for the School of STEM: See

<https://www.uwb.edu/stem/about/stem-policies> for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity. Student Support Services, Wondering How to Address Faculty? etc. (P.S. I prefer to be addressed as 'Professor Pisan')

Course Material: Lecture notes and other material will be posted to Canvas under "Files" the week after the lecture.

Topics

Week 1: Introduction to C++

Week 2: Variables / Assignments

Week 3: Branches - Exam 1: Thursday Oct 10th

Week 4: Loops

Week 5: Arrays / Vectors -- No Class Thursday Oct 24th, exercises/video will be assigned

Week 6: User-Defined Functions -- Exam 2: Thursday 31st

Week 7: Objects and Classes

Week 8: Pointers

Week 9: Streams -- Exam 3: Thursday Nov 21st

Week 10: Inheritance -- No Class Thursday Nov 28th, Thanksgiving

Week 11: Review

Week 12: Final exam: Thursday, Dec 12th, 5:45pm